MIL-E-1/1593B
27 June 1983
SUPERSEDING
MIL-E-1/1593A
10 January 1969

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 8233

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Pentode, sharp cutoff

Outline --- See figure 1

Base --- E9-23

Envelope --- T9

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	a	g3	g2	h	h	g3	k	g1	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ebb	Eb	Ecc2	Ec2	Ec3	Ecc1	Ec1	egk
Unit:	V	V dc	V dc	V dc	V dc	V dc	V dc	V dc	v
Maximum:	6.6	400	200	350	175	---	---	0, -55	0
Minimum:	6.0	---	---	---	---	---	---	---	-110
TEST CONDITIONS:	6.3	136	---	136	---	0	+8	---	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ehk	Rk	Rg1	Ik	Pg2	Pp	TE	Alt
Unit:	v	Ohms	Meg	mA dc	W	W	°C	ft
Maximum:	200	---	0.125	75	1.5	10	180	60,000
Minimum:	---	---	---	---	---	---	---	---
TEST CONDITIONS:	0	200	---	---	---	---	---	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1301	Heater current			0.65	II	If	540	660	mA dc
1336	Heater-cathode leakage			0.65	II	Ihk	---	15	µA dc
1266	Total grid current			0.65	II	Ic1	0	-1.0	µA dc
1261	Electrode voltage (grid 1)	1		0.65	II	Ec1	2.3	3.7	V dc
1256	Electrode current (screen)			0.65	II	Ic2	---	7.0	mA dc
1306	Transconductance (1)			0.65	II	Sm	38,000	52,000	µmhos
1256	Electrode current (anode)		Ec1 = -6 V dc; Eb = 125 V dc; Ec2 = 125 V dc	0.65	II	Ib	---	1.0	mA dc
1201	Short and discontinuity detection		egk = 110 v (max)	0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes		E (g1 to all) = -50 V dc E (a to all) = -200 V dc	2.5	I	---	---	---	---
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm/Ef	---	10	%
1316	Amplification factor	2		6.5	S3	Mu	26	34	---
1266	Grid emission	3	Ef = 7.5 V; Ecc1 = -10 V dc; Rg1 = 0.1 Meg	2.5	I	Ic1	0	-2.0	µA dc
1331	Direct-interelectrode capacitance		No shield	6.5	Code E	Cin	15	21	pF
						Cout	3.6	4.4	pF
						Cgp	---	0.15	pF
1002	Barometric pressure, reduced		Pressure = 55 ±5 mm Hg; voltage = 300 V ac	6.5	See note 4	---	---	---	---
1031	High-frequency vibration		Rp = 2,000 ohms; Rk = 56 ohms, unbypassed; Rg2 = 20,000 ohms; Cg2 = 25 µF; Ebb = Ecc2 = 225 V dc; Ec1 = 0; F = 50 Hz	2.5	Code H	Ep	---	50	mV ac
1041	Shock		450 G; no voltages applied	6.5	See note 4	---	---	---	---

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 2 - continued								
1031	Vibration-fatigue			6.5	See note 4	---	---	---	---
---	Shock and vibration-fatigue-test end points:								
1031	High-frequency vibration		Rp = 2,000 ohms; Rk = 56 ohms, unbypassed; Rg2 = 20,000 ohms; Cg2 = 25 µF; Ebb = Ecc2 = 225 V dc; Ec1 = 0; F = 50 Hz	---	---	Ep	---	100	mV ac
1336	Heater-cathode leakage			---	---	Ihk	---	25	µA dc
1266	Total grid current			---	---	Ic1	0	-2.0	µA dc
1306	Change in transconductance (1) of individual tubes			---	---	ΔSmt	---	20	%
2126	Envelope strain			2.5	1	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3						
1506	Heater-cycling life	5	Ef = 7.5 V; Ehk = 140 V ac; Ec1 = Ec2 = Eb = 0	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	25	µA dc
1516	Stability life	5	Ehk = -200 V dc; Rg1 = 0.1 Meg; TA = room	---	---	---	---
---	Stability life-test end point						
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	10	%
1501	Intermittent life	6	Ehk = -200 V dc; Rg1 = 0.1 Meg	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3 - Continued						
---	Intermittent life-test end points (500 hours)	7					
1266	Total grid current			Ic1	0	-2.0	µA dc
1301	Heater current			If	530	670	mA
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	20	%
1306	Transconductance (2)		Ef = 5.7 V	ΔSm Ef	---	15	%
1336	Heater-cathode leakage			Ihk	---	30	µA dc
1211	Insulation of electrodes		E (g1 to all) = -50 V dc E (a to all) = -200 V dc	---	---	---	---
1306	Transconductance (1), average change			Avg ΔSm t	---	15	%
---	Intermittent life-test end points (1,000 hours)	8					
1266	Total grid current			Ic1	0	-2.0	µA dc
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	25	%
1336	Heater-cathode leakage			Ihk	---	30	µA dc
1211	Insulation of electrodes		E (g1 to all) = -50 V dc E (a to all) = -200 V dc	---	---	---	---

NOTES:

1. This voltage shall be read with a high-impedance voltmeter connected between grid and cathode.
2. The screen grid shall be utilized as the anode (output is in screen grid circuit). Anode voltage = 136 V dc.
3. Prior to this test, tubes shall be preheated 5 minutes at the conditions indicated below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for grid emission test.

Ef	Ebb	Ecc2	Ecc1	Rg1	Rk
V	V dc	V dc	V dc	Meg	Ohms
7.5	136	136	+8	0.125	200

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NOTES - Continued.

4. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

5. The AQL shall be 1.0, inspection level S3.

6. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.

7. The allowable defectives per characteristic shall be not greater than 1 for the first sample and shall be not greater than 3 for combined samples.

8. The allowable defectives per characteristic shall be not greater than 2 for the first sample and shall be not greater than 5 for combined samples.

9. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

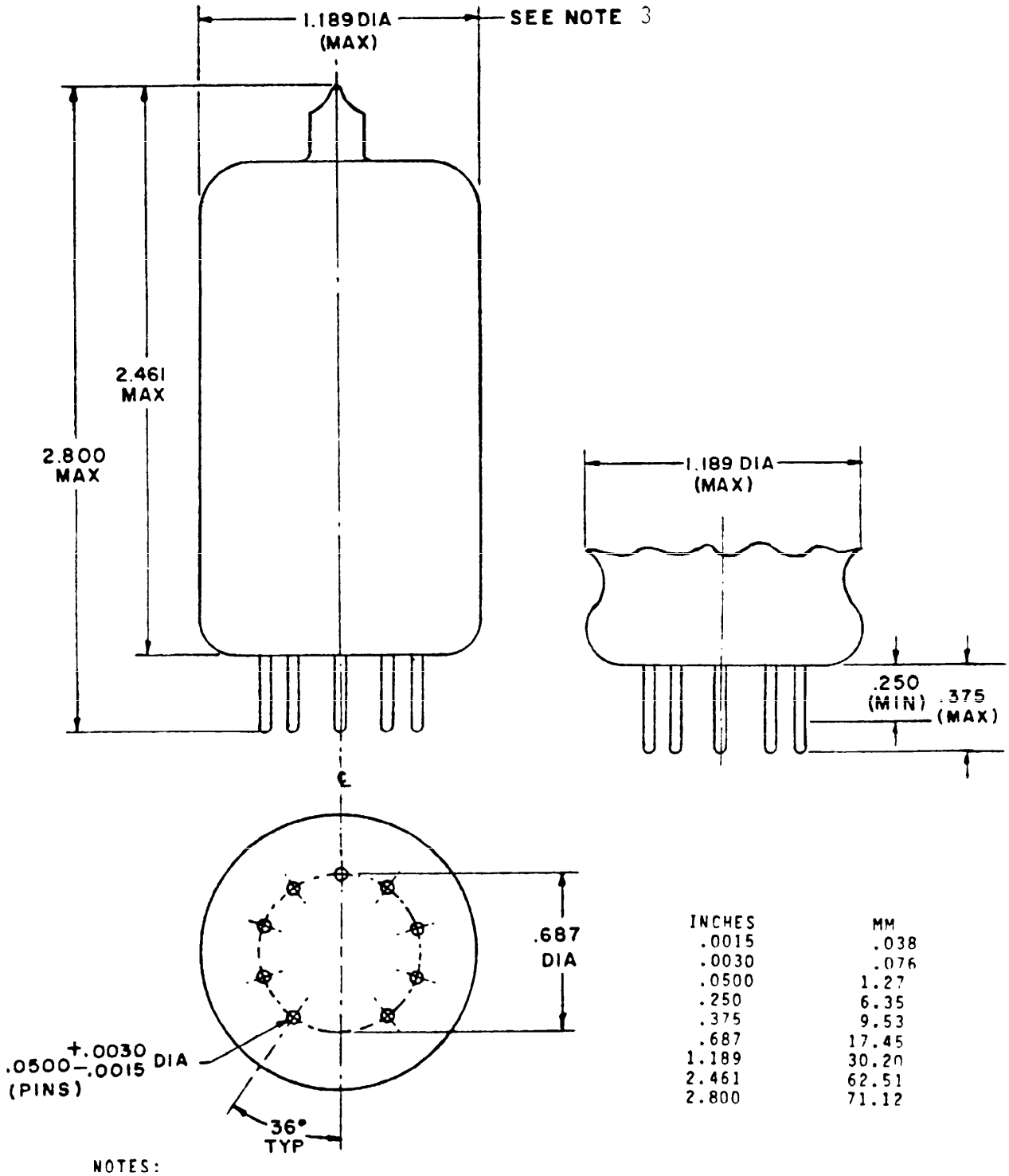
User activities:
Army - ME
Navy - AS, OS, MC, CG, SH
Air Force - 11, 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5960-3304)

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INCHES	MM
.0015	.038
.0030	.076
.0500	1.27
.250	6.35
.375	9.53
.687	17.45
1.189	30.20
2.461	62.51
2.800	71.12

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based on 1 inch = 25.4 mm.
3. The minimum measurement should be made in the area included between .375 (9.53 mm) inch from the bottom of bulb to the top of bulb body.

FIGURE 1. Outline drawing of electron tube type 8233.

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